

The obligation to succeed when it matters the most—The influence of skill and pressure on the success in football penalty kicks

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ABSTRACT

In order to gain a more comprehensive understanding of the influence of skill and pressure on the success in football penalty kicks, we analyzed 1711 penalties taken over a 15-year period in major international tournaments. We conducted a multiple correspondence analysis in order to reduce six variables that are associated with skill and pressure to two dimensions that reflect our target concepts. Then, we used these two factors as independent variables in a logistic regression and fit different models using three binary dependent variables. The results show that high situational pressure significantly increases the probability of missing the goal entirely by about 6%, independent of the player's skill level. The probability that the goalkeeper saves a penalty significantly decreases by roughly 4% when a highly skilled player takes the shot. In general, high situational pressure decreases the probability of scoring a penalty kick. Furthermore, the probability to score a penalty increases if a highly skilled player takes the kick which indicates that a high skill level can act as a kind of buffer against debilitating effects caused by performance pressure.

1. Introduction

Penalty kicks in football have frequently been described as a “lottery”, as quotes from prominent coaches like Luiz Felipe Scolari or Fabio Capello show (Wood, Jordet, & Wilson, 2015). However, in contrast to this repeatedly announced opinion, a growing body of empirical research has shown that many other factors besides luck have a systematic influence on the likelihood of scoring a penalty kick (for overviews, see Memmert, Hüttermann, Hagemann, Löffing, & Strauss, 2013; Memmert & Noel, 2020; Wood et al., 2015). The existing research often targets one factor that arguably has a major influence on the success in football penalties, namely the amount of pressure associated with the kick due to its contextual importance (e.g., Brimmell, Parker, Furley, & Moore, 2018; Dohmen, 2008; Ellis & Ward, 2022; Furley, Dicks, & Jordet, 2020; Jordet, Hartman, & Jelle Vuijk, 2012; Navarro et al., 2012; Navia, van der Kamp, Avilés, & Aceituno, 2019; Wood, 2010). Another factor that we expect to play a crucial role in penalty kicks had not yet received much attention in the current literature: The skill level of the players involved in the penalty kick situation.

Scoring a penalty kick seems to be quite easy, especially for a professional football player: The shot is taken from a spot that is only twelve

yards (11 m) away from the goal and no other field player is allowed to interfere in the process. In addition, the target area seems quite big as it measures 24 ft (7.32 m) wide by 8 ft (2.44 m) high, which amounts to approximately 192 ft² (18 m²) for the penalty taker to hit. The only obstacle is the opposing goalkeeper for whom it should be extremely difficult to successfully guard the net if the penalty is taken with enough force in the right direction. A typical penalty that is shot with a speed of about 20 m/s reaches the goal-line after about 500–600 ms (Franks & Harvey, 1997; Kuhn, 1988; van der Kamp, Dicks, Navia, & Noël, 2018; Wood et al., 2015). Therefore, a well-executed penalty kick to the distal areas of the goal is almost impossible to save for a goalkeeper due to constraints on their reaction time (Bar-Eli & Azar, 2009; Bowtell, King, & Pain, 2009; Dicks, Davids, & Button, 2010a; Franks & Harvey, 1997; Tsai, 2005; van der Kamp et al., 2018; Wood et al., 2015). In addition, research has shown that penalties that are kicked into the upper third of the goal are hardly ever saved by the goalkeeper, but surprisingly, only about 15% of penalties are kicked into this area (Almeida, Volossovitch, & Duarte, 2016; Bar-Eli & Azar, 2009). A possible explanation for a lack of penalties kicked into the upper corners might be that players are instructed to kick low and close to the goalpost, as aiming for the upper zones is considered riskier due to an increased difficulty on a technical

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level (Bar-Eli & Azar, 2009; Navia et al., 2019). In bold contrast, other research has shown that a total of about 25% of penalties on a high professional level are either saved by the goalkeeper or miss the goal entirely (Brinkschulte, Furley, & Memmert, 2020; Brinkschulte, Furley, Klemp, & Memmert, 2021; Dalton, Guillon, & Naroo, 2015; McGarry & Franks, 2000). In this regard, past research has shown that many penalties are not kicked very accurately, meaning that the ball does not enter the goal in the (supposedly) optimal areas close to the posts (see Bar-Eli & Azar, 2009; Bar-Eli & Friedman, 1988; Miller, 1996 for more detailed information). Taken together, this research does not only reflect the impact of situational pressure on the success in penalty kicks but, moreover, it certainly indicates that a high skill level should theoretically have a facilitating effect on the ability to score a penalty as it enables the player to take the kick in such a way that it is almost impossible for the goalkeeper to save (e.g., with appropriate speed into the upper third of the goal, see Bar-Eli & Azar, 2009; van der Kamp et al., 2018).

Surprisingly, the influence of a player's skill level on the success in football penalties has been rather neglected in the scientific literature. One example is a study conducted by Krumer (2020) that compared the success in 586 penalty shootouts between teams that play in different divisions. The results indicate that a difference of only one division increases the chance of winning for the team from the higher division by 8%. The author concludes that the skill level of the players is of major importance in high-pressure situations, even in a simple mechanical task such as the football penalty kick (Krumer, 2020). This is a promising finding that may support the hypothesis of our study. However, Krumer (2020) only considered the mere outcome of whole penalty shootouts (in contrast to individual penalty kicks) and the only operationalization of skill was the categorization of teams from different divisions. A second example is a recent experimental study by Brimmell, Edwards, Smith, and Vaughan (2021) that considered the influence of skill on the success in penalty kicks. Their research mainly focused on the relationship between executive function, visual attention, and performance in penalty kicks under high versus low pressure, however, they also controlled for expertise. Here, an expertise score was calculated (based on Hagyard, Brimmell, Edwards, & Vaughan, 2021; Swann, Moran, & Piggott, 2015; Vaughan & Edwards, 2020) that was used to assign an expertise level to the participants of the study. The results show that the performance significantly increased with a higher expertise level, which supports our assumption that a player's skill level has an influence on their ability to score a penalty kick.

Research conducted by Jordet, Hartman, Visscher, and Lemmink (2007) that aimed to investigate the influence of skill (amongst other variables) on penalty kick success provides a counterexample. Here, skill was operationalized via the playing position (attackers = high level of skill in scoring penalties) and the player's age was used as a proxy for experience (high age = high level of skill). In regard to the playing position, the authors assumed that players whose most important task is to score goals in-game will also be better at scoring from the penalty mark. The results revealed that there was no significant association between the proxies used to quantify the skill level and the penalty kick success rates of the players. In this respect, the authors concluded that a player's skill level is not predictive of the outcome of a penalty kick. However, although the results may be representative for the proxies used (namely, playing position and age), we do not necessarily consider these operationalizations of skill as methodologically sufficient. Especially the playing position should not be considered an appropriate proxy for skill because the situational and technical requirements of scoring a goal in-game are substantially different from scoring a penalty kick. On a situational level, in-game goals are scored in a highly dynamic environment while penalty kicks are taken in a rather static environment. On a technical level, goals can be scored in basically countless different ways in-game while penalty kicks are taken in a rather standardized procedure. Based on these considerations, we decided to return to the question of the relationship between skill and penalty kick success rates in football by using indicators that we believe reflect skill in a more

applicable way.

The study by Jordet et al. (2007) did not find that a player's on-field position or age influences penalty kick success, but it did show an impact of the importance of the kick on the success rate. The authors argue that the psychological pressure increases along with the importance of the kick, which leads to a lower chance of scoring it. In follow-up research, Jordet and colleagues have published several studies on how pressure impacts the players who take a penalty kick. Two studies (Jordet, Hartman, & Sigmundstad, 2009; Jordet & Hartman, 2008) provided evidence that the behavior of players who were taking a particularly important penalty kick was influenced in a negative manner which subsequently led to a lower likelihood of scoring the penalty. Based on these results, the authors concluded that high levels of pressure cause penalty takers to show "escapist behaviors" that lead them to try to get the situation "over and done with" as quickly as possible. This behavior contributes to negative performance, in addition to the debilitating effects of performance anxiety on penalty success. In this regard, a recent experimental study by Navia et al. (2019) that investigated the success in penalty kicks under pressure focused on the players degree of control in decision-making. The results confirmed that pressure in penalty kicks increases the players' level of cognitive and somatic anxiety levels. Also, players with increased external control scored less penalty kicks and took less time preparing and performing the run-up—meaning that Navia et al. (2019) were able to experimentally replicate the performance deteriorations and "escapist behaviors" shown by players under high-pressure described by Jordet et al. (2007). Furthermore, another experimental study that shows comparable results was recently carried out by Ellis and Ward (2022). They examined the physiological and psychophysical responses of professional football players as well as their thoughts, feelings, and experiences in penalty kicks under high and low pressure. The authors applied a high-pressure penalty kick protocol in order to simulate realistic high-pressure situations. Players experienced increased perceived pressure, cognitive anxiety, and respiration rate while shooting accuracy and self-confidence decreased under high pressure.

While there is a lack of specific literature on the combined influence of skill and pressure on the success in penalty kicks, there are several studies within the field of economics on high-pressure situations in which individuals need to perform at particularly high levels. Pertinent to the present investigation, this research indicates that a person's skill level and the contextual pressure do not seem to operate as independent factors in influencing success in high-pressure situations. For example, empirical results of a number of studies suggest that "a highly skilled agent performs best when it matters most" (Cohen-Zada, Krumer, & Shapir, 2018; Ehrenberg & Bognanno, 1990; González-Díaz, Gossner, & Rogers, 2012; Iqbal & Krumer, 2019; Jetter & Walker, 2015; Krumer, 2020; Lazear, 2000; Rosen, 1986). Several of these studies were able to show that the situational performance of individuals with great skill can improve considerably when a lot is at stake (Cohen-Zada et al., 2018; González-Díaz et al., 2012; Iqbal & Krumer, 2019; Jetter & Walker, 2015; Lazear, 2000; Rosen, 1986). In a sports context, González-Díaz et al. (2012) demonstrated that top tennis players show a higher "critical ability", which means that they perform better than others in decisive high-pressure situations (e.g., important matches in prestigious international tournaments such as the US Open). These players are more likely to win the crucial points in a match (e.g., break points), which consequently has a significant long-term impact on their overall career success.

The reviewed studies of Jordet and colleagues (Jordet & Hartman, 2008; Jordet et al., 2007; Jordet et al., 2009; Jordet et al., 2012) suggest that pressure would influence performance, but that the skill level would not. The results of Krumer (2020) and Brimmell et al. (2021) indicate that a high skill level has a positive impact on the outcome of penalty shootouts, while a further study even finds tentative evidence that football players might even be more inclined to perform worse on penalties when they have a particularly high skill level (Jordet, 2009a).

The reviewed research from the field of economics, on the other hand, suggests that the interaction of a person's skill level and the situational pressure can influence success in sports. Some studies indicate that a high skill level might act as a kind of buffer against debilitating effects caused by performance pressure (Cohen-Zada et al., 2018; Ehrenberg & Bognanno, 1990; Iqbal & Krumer, 2019; Jetter & Walker, 2015; Lazear, 2000; Rosen, 1986), while others show that highly skilled athletes might perform even better under high pressure (González-Díaz et al., 2012).

As the existing literature does not allow to derive a straight-forward hypothesis on the influence of skill and pressure on the performance in football penalties, we examined extensive archival data in an explorative manner. In order to gain a more comprehensive understanding of the influence of various variables that are associated with skill and pressure on the success in penalty kicks, we systematically analyzed a large number of penalty kicks taken in four major international tournaments (FIFA World Cup, UEFA European Championship, UEFA Champions League, UEFA Europa League).

2. Method

2.1. Data collection

Our methodological approach in regard to the data collection is based on the one of Brinkschulte et al. (2020, 2021). First, we collected and coded comprehensive data on all 1711 penalty kicks taken in the FIFA World Cup, UEFA European Championship, UEFA Champions League, and UEFA Europa League from 2005/06 to 2019/20. The 2005/06 season marks our starting point because we included market values and betting odds as independent variables in our data set and there is no reliable information available about these variables prior to that season.

Our dataset includes specific information on each individual penalty kick taken in the mentioned competitions and time period. First, we researched the name of the player who took a shot and the name of the goalkeeper who faced it as well as the names of the teams they played for. We collected the pre-game betting odds for the player's team and the goalkeeper's team as well as for a tie in order to be able to assess the *team quality* (i.e., the stronger and weaker team in a particular match). In addition, we researched the player's and goalkeeper's *market values* at the exact point in time the penalty kick was taken to operationalize player and goalkeeper quality. Furthermore, we added information on the *type of competition* (i.e., whether the penalty kick was taken for a player's club during the UEFA Champions League/UEFA Europa League or for a player's national team during the FIFA World Cup/UEFA European Championship) and the *tournament phase* (i.e., at which stage of the tournament the penalty kick was taken). As all four tournaments consist of a group phase and a knock-out stage, we distinguished between the two possible *types of penalty kicks* (i.e., in-game or shootout). We determined the team quality, player market value, goalkeeper market value, type of competition, tournament phase, and type of penalty kick as our six independent variables. All data points were collected from several web-based sources (e.g., transfermarkt.de, fifa.com, uefa.com, kicker.de, soccerstats.com, worldfootball.net) and the information was double-checked using multiple of these sources for each penalty kick. We operationalized our binary dependent variable *success vs. failure* in three different ways: (1) the penalty kick was *scored vs. not scored* (regardless of whether it was saved by the goalkeeper or missed the goal entirely), (2) the penalty was *shot on target* (regardless of whether it was scored or saved by the goalkeeper) *vs. missed the goal entirely*, and (3) the ball was *not saved by the goalkeeper vs. saved by the goalkeeper* (solely the 1574 penalties that were shot on target).

Player and goalkeeper market values were collected using the web-based platform *transfermarkt.com* as this website offers appropriate market value estimations for professional football players. The basis for these estimations is an interconnected community of users who observe and discuss relevant public information to decide on a realistic market

value for each player. The final decision is made by "a few merited community members" (Herm, Callsen-Bracker, & Kreis, 2014), while the possibility to propose and discuss market values is in principle open to any user. For a more detailed discussion on the process of the market value estimation, we refer to the work of Herm et al. (2014) and Peeters (2018). Due to their good availability and the easy access, market values are used as a common resource in the field of football analysis. Here, *transfermarkt.com* has proven to be a specifically useful and reliable source for researchers (Herm et al., 2014; Peeters, 2018; Prockl & Frick, 2018). The accuracy of *transfermarkt.com* has been demonstrated as market values have been shown to be a good predictor of actual transfer fees (Herm et al., 2014) and match outcomes (Peeters, 2018), without being subject to a personal bias of community members in overestimating players of their favorite clubs (Peeters, 2018). Player and goalkeeper market values are constantly developing over time, reflecting the financial development of the football business. Therefore, we had to take market value inflation into account as we evaluated a relatively long time period of 15 seasons. For the considered seasons, we researched all player market values as well as the roster sizes for all teams in the five biggest European football leagues (English Premier League, Spanish La Liga, German Bundesliga, Italian Serie A, French Ligue 1) and then calculated the average player market value for each season. Figure 1 illustrates the market value development over the last 15 seasons from 2005/06 to 2019/20. We used the most recent season (2019/20) as a benchmark and adjusted the market values in our dataset to reflect the level of that season. For example, if the average player market value of a season was 50% of the average player market value of the 2019/20 season, we doubled the market value of each player from that particular season to ensure the best possible comparability (we excluded three penalty kicks taken by goalkeepers from the dataset as the market value of a goalkeeper may have a limited comparability to the market value of on-field players). We combined the data from the FIFA World Cups and the UEFA European Championships with the data of the season preceding the tournament, e.g., the 2014 World Cup was combined with the 2013/14 season.

The pre-game betting odds for each match in our dataset were obtained from *football-data.co.uk* and *oddsportal.com*. Betting odds provided by bookmakers are directly related to the probability of a team winning, losing, or drawing a match. They can be easily translated into percental forecasts by calculating the inverse of the betting odds and extracting the bookmaker margin. This approach has been proven to be an excellent predictor for the outcome of football matches (Forrest, Goddard, & Simmons, 2005; Hvattum & Arntzen, 2010; Spann & Skiera, 2009; Wunderlich & Memmert, 2018). Besides being affected by the qualities of the teams, the home advantage has a notable impact on betting odds (for an overview see Pollard, 2008). As we are interested in the unbiased team quality, we excluded the effects of home advantage by calculating the difference in winning probabilities in line with the approach of Wunderlich and colleagues (2020). Then, we derived a binary variable (*stronger vs. weaker*) for the two teams of each match.

2.2. Data analysis

We assume that the football-specific skill of a player and the situation-dependent psychological pressure are two major factors that can influence the success in penalty kicks. Unfortunately, the two concepts of *skill* and *pressure* cannot be measured directly as it is not possible to assign them a naturally observable value in our archival data. Therefore, we use six independent variables of which two can be considered a reasonable reflection of the skill of a player (the team quality and the player market value as a proxy for his quality) and four can be argued to have a significant impact on pressure (the market value of the opposing goalkeeper as a proxy for his quality, the type of competition, the tournament phase, and the type of penalty kick). As a first step, we therefore conducted χ^2 -tests in order to analyze the individual influence that each of these six independent variables has on the

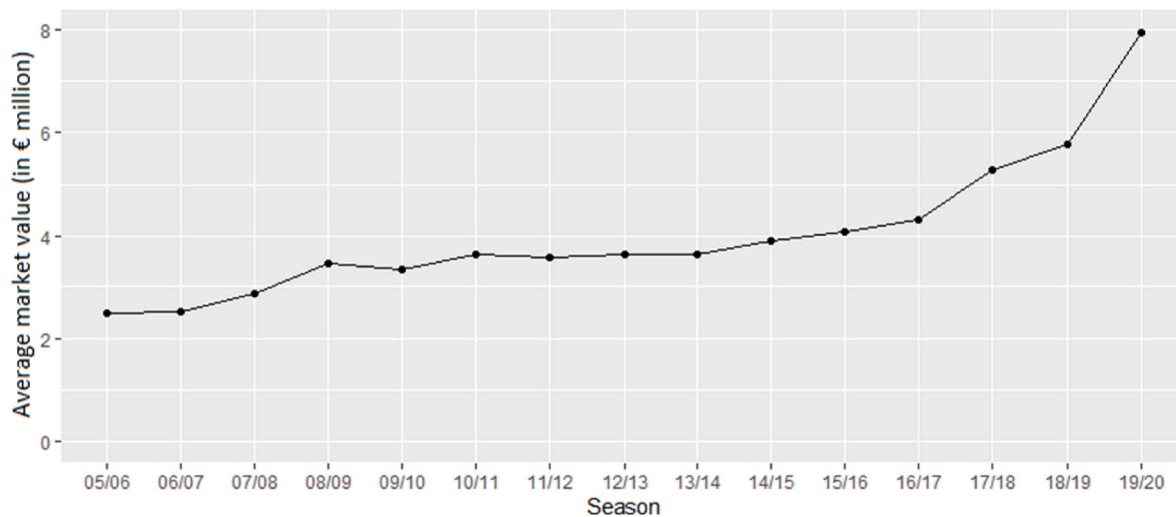


Figure 1. Development of the average player market values per season in five major European football leagues (Premier League, La Liga, Bundesliga, Serie A, Ligue 1) over 15 seasons (2005/06 to 2019/20). Market values are displayed in € million.

probability that the penalty kicks in our dataset were scored, missed the goal entirely, or were saved by the goalkeeper. As a second step, we conducted Spearman correlations and report the correlation coefficients (r_s) in the results section in order to make potential relationships between our six independent variables visible.

However, there are two major issues when aiming at a statistically clear investigation of the role that pressure and skill can potentially play in penalty kicks. First, some of the variables might affect both factors at the same time such as the market value of the player. On the one hand, it can be considered reasonable to argue that players with a high market value also have a higher skill level and should therefore be more successful in penalty kicks. On the other hand, the penalty-taker could experience a higher degree of self-inflicted pressure because he might think that it is expected from him to perform better than players with a lower skill level (for a related discussion, see [Jordet, 2009a](#)). Second, some of these variables might be naturally intercorrelated: For example, a team that has players with market values that are higher than the market values of the players of the opposing team will also be the bookmaker's favorite in regard to the betting odds. Another example is that penalty shootouts are naturally related to later tournament phases because they simply do not occur in the group stage.

For these reasons, we aim to reduce the six-dimensional data in our dataset to the two factors *skill* and *pressure*. One possibility to reduce multi-dimensional data is to develop an equation based on a theoretical model. In our case, the procedure would be to determine two factors based on our six independent variables, purely driven by theoretical considerations. We developed such a theoretical model based on the following assumptions: First, we assume that taking penalties in a shootout, taking penalties for a national team, and taking penalties in a late tournament phase are major indicators of pressure. Second, we assume that a high general quality of the team as well as a high individual market value of the player are major indicators of skill. Third, we assume that a high goalkeeper quality and reputation positively contributes to pressure and negatively contributes to skill (as skill is measured in comparison to the opponent). To make sure that market values, as all other variables, contribute a maximum of 1 to the scores, the market values are multiplied by a factor of 0.5 and the market value of the goalkeeper is additionally split up to both components. Taking these theoretical considerations into account, our scores for *skill* and *pressure* are calculated as follows:

$$\begin{aligned} \text{skill} = & \text{team quality} \{0 = \text{weaker}, 1 = \text{stronger}\} \\ & + 0.5 \cdot \text{player market value} \{0 = \text{low}, 1 = \text{middle}, 2 = \text{high}\} \\ & - 0.25 \cdot \text{goalkeeper market value} \{0 = \text{low}, 1 = \text{middle}, 2 = \text{high}\} \end{aligned}$$

$$\begin{aligned} \text{pressure} = & \text{penalty type} \{0 = \text{ingame}, 1 = \text{shootout}\} \\ & + \text{competition type} \{0 = \text{club team}, 1 = \text{national team}\} \\ & + \text{tournament phase} \{0 = \text{before round of 8}, 1 = \text{from round of 8}\} \\ & + 0.25 \cdot \text{goalkeeper market value} \{0 = \text{low}, 1 = \text{middle}, 2 = \text{high}\} \end{aligned}$$

The advantage of such a theoretical model is that we can deliberately chose variables that can be argued to be related to *pressure* and *skill* from a theoretical standpoint. The disadvantage of this model is, however, that it arguably involves a significant degree of subjectivity while no objective evidence for the correctness of the variables and weighting of variables exists.

Statistically reducing data is a use case for principal component analysis (PCA; [O'Donoghue, 2008](#)) or, in case of categorical data, for multiple correspondence analysis (MCA; [Kamalja & Khangar, 2017](#)). In contrast to a theoretical model, by using PCA and MCA, it is possible to reduce dimensions on a statistical basis rather than on a content basis. When using PCA or MCA, the procedures themselves neither know anything about the concepts of *pressure* and *skill* nor do they consider the values of the dependent variable. Therefore, it is not clear from the outset that such a method would actually yield dimensions that adequately reflect *skill* and *pressure*. However, based on the theoretical considerations made in regard to our theoretical model, it can be assumed that the values obtained from a PCA or MCA can adequately reflect the idea behind our target variables *skill* and *pressure*.

Based on all the considerations above and the fact that our dataset contains inherently categorical data, we decided to perform and report an MCA as our main analysis. As a robustness check, both the results of our theoretical model and of a PCA will be made available as supplementary online material. After reducing our six variables to two factors by using an MCA, we used these two factors as independent variables in a logistic regression model. We fit different models using our three binary dependent variables, namely *scored vs. not scored*, *shot on target vs. missed the goal entirely*, and *not saved by the goalkeeper vs. saved by the goalkeeper*. We performed the analyses using the statistical software programs IBM SPSS Statistics ([IBM Corp., 2021](#)) and R ([R Development Core Team, 2021](#)). When evaluating the results, we distinguish between significance levels of 5%, 1%, and 0.1%.

3. Results

3.1. General descriptive results and χ^2 -Tests

Our dataset contains a total of 1711 penalty kicks of which 1272 (74.3%) were scored and 439 (25.7%) were not scored. Considering all

attempts, 137 (8.0%) penalty kicks missed the goal entirely and 1574 (92.0%) were shot on target (regardless of whether it was scored or saved by the goalkeeper). Furthermore, as we were also interested in goalkeeper success, we separately evaluated the attempts that were shot on target because penalty kicks that miss the goal entirely do not give the goalkeeper a chance to save it. Of the 1574 penalties that were shot on target, the goalkeepers saved 302 (19.2%) which means that 1272 (80.8%) of these attempts were scored. [Table 1](#) summarizes the percentages of the penalty kicks that were scored, missed the goal entirely, and were saved by the goalkeeper in regard to our six independent variables individually. In addition, we present p -values for the conducted χ^2 -tests for each of the variable combinations.

3.2. Spearman correlations

The results of the conducted Spearman correlations show that the majority of our independent variables are intercorrelated. With regard to the situation the penalty kick was taken in, the *type of competition*, the *tournament phase*, and the *type of penalty kick* are significantly correlated (*tournament phase* and *type of penalty kick*, $r_s(1709) = 0.48, p < .001$; *type of competition* and *type of penalty kick*, $r_s(1709) = 0.42, p < .001$; *tournament phase* and *type of competition*, $r_s(1709) = 0.30, p < .001$). With regard to the quality of teams, on-field players, and goalkeepers the variable *team quality* was found to be positively correlated to the *player market value* ($r_s(1709) = 0.33, p < .001$) and negatively correlated to the *goalkeeper market value* ($r_s(1709) = -0.29, p < .001$) while the *player*

market value and the *goalkeeper market value* are positively correlated ($r_s(1709) = 0.17, p < .001$). Moreover, the *tournament phase* is positively correlated to the *player market value* ($r_s(1709) = 0.20, p < .001$) and to the *goalkeeper market value* ($r_s(1709) = 0.28, p < .001$). Finally, the *type of penalty kick* and the *goalkeeper market value* are positively correlated ($r_s(1709) = 0.27, p < .001$). All remaining correlations fall below an absolute value of $r_s = 0.15$.

3.3. Multiple correspondence analysis

The results of the MCA can be found in [Table 2](#). Dimension 1 is strongly associated with the *team quality* and the *player market value*

Table 2

Results of the MCA analysis, illustrating the relationship between our six original independent variables and the two dimensions. Discrimination measures exceeding a value of 0.1 are marked bold.

Original Variable	Dimension 1	Dimension 2	Average
Team Quality	0.670	0.082	0.376
Player Market Value	0.626	0.052	0.339
Goalkeeper Market Value	0.029	0.355	0.192
Type of Competition	0.006	0.404	0.205
Tournament Phase	0.023	0.566	0.295
Type of Penalty Kick	0.000	0.610	0.305
Total (Eigenvalue)	1.354	2.069	1.712
Variance Explained	22,566	34,486	28,526

Table 1

Success and failure ratios of penalty kicks allocated to our different independent variable categories. The p -values represent the results of conducted χ^2 -tests. The accumulated percentages per row do not add up to 100% because for our dependent variable *saved by the goalkeeper*, we only considered the 1574 penalty kicks that were actually shot on target (i.e., we excluded the ones that missed the goal entirely). For the dependent variables scored and missed the goal entirely, all 1711 penalty kicks were analyzed.

Independent Variables		Dependent Variables					
		Scored		Missed the Goal Entirely		Saved by the Goalkeeper	
		Ratio	p	Ratio	p	Ratio	p
Team Quality (Skill)	Weaker	71.2%	.017*	9.6%	.044*	21.2%	.110
	Stronger	76.4%		6.9%		17.9%	
Player Market Value (Skill)	Low (< €10M)	73.2%	.776	7.1%	.207	21.2%	.249
	Middle (< €35M)	74.8%		7.3%		19.3%	
	High (≥ €35M)	74.9%		9.6%		17.1%	
Goalkeeper Market Value (Skill / Pressure)	Low (< €4M)	75.8%	.400	6.1%	.023*	19.3%	.996
	Middle (< €15M)	74.6%		7.8%		19.1%	
	High (≥ €15M)	72.3%		10.5%		19.2%	
Type of Competition (Pressure)	Club Team	75.0%	.169	6.1%	< .001***	20.2%	.020*
	National Team	71.1%		17.5%		13.8%	
Tournament Phase (Pressure)	Before Round of 8	75.1%	.179	7.1%	.006**	19.2%	.958
	From Round of 8	71.5%		11.6%		19.1%	
Type of Penalty Kick (Pressure)	In-Game	75.0%	.286	6.7%	< .001***	19.6%	.468
	Shootout	72.5%		11.7%		18.0%	

Note. * $p < .05$. ** $p < .01$. *** $p < .001$.

while dimension 2 is strongly associated with the *goalkeeper market value*, the *type of competition*, the *tournament phase*, and the *type of penalty kick*. As such, dimension 1 appears to be related to skill, while dimension 2 appears to be related to pressure. In the following, we will therefore refer to the two dimensions as *skill* and *pressure* respectively.

Resulting from the calculations of the MCA, we obtained both a *skill* and a *pressure* score for each penalty kick in our dataset. These scores reflect skill and pressure as standardized values with a mean of 0 and standard deviation of 1 (which explains the existence of negative values). In this context, it must be mentioned that a low skill score does by no means reflect low football-specific abilities of a player (or goalkeeper). We only included the highest international competitions and all athletes in our dataset are considered exceptional players (and goalkeepers) compared to the general population of football players. This means our skill score is solely a relative value that allows us to put the data points from our dataset in relation to each other. A descriptive two-dimensional representation of all our data points can be found in Figure 2. Here, the x-axis represents *pressure* and the y-axis represents *skill*. As an example of the different combinations of skill and situational pressure, we highlighted four particular penalty kicks in the figure:

The circle in the bottom left reflects low situational pressure (−0.87) and a comparably low skill (−1.45). It represents an in-game penalty kick taken by Ole Martin Årst (adjusted market value of €1.6 million) against goalkeeper Vladimir Stojković (adjusted market value of €2.2 million) in a 2005/06 UEFA Europa League group match between Tromsø IL and Red Star Belgrade. According to the betting odds, Årst's team Tromsø IL was the weaker team in this match.

The circle in the bottom right reflects high situational pressure (2.42) but a comparably low skill (−1.32). It represents a penalty kick taken by Emanuele Giaccherini (adjusted market value of €7.8 million) against goalkeeper Manuel Neuer (adjusted market value of €87.5 million) in the penalty shootout of the quarter final of the 2016 UEFA European Championship between Italy and Germany. According to the betting odds, Giaccherini's team Italy was the weaker team in this match.

The circle in the top left reflects low situational pressure (−0.87) but a comparably high skill (1.23). It represents an in-game penalty kick taken by Robert Pirès (adjusted market value of €43.0 million) against goalkeeper Eldin Jakupović (adjusted market value of €2.4 million) in a 2005/06 UEFA Champions League group match between FC Arsenal London and FC Thun. According to the betting odds, Pirès' team FC Arsenal London was the stronger team in this match.

The circle in the top right reflects high situational pressure (2.41) and a comparably high skill (1.36). It represents a penalty kick taken by David Villa (adjusted market value of €78.2 million) against goalkeeper Gianluigi Buffon (adjusted market value of €83.8 million) in the penalty shootout of the quarter final of 2008 UEFA European Championship

between Spain and Italy. According to the betting odds, Villa's team Spain was the stronger team in this match.

In order to investigate the relationship between the two dimensions *skill* and *pressure* (obtained from the conducted MCA) and our three dependent variables *scored*, *missed the goal entirely*, and *saved by the goalkeeper*, we conducted binary logistic regressions. Table 3 illustrates the results of the three logistic regression models.

In order to exclude the possibility that the results are only an artifact of our MCA-based definition of *pressure* and *skill*, we performed robustness tests using PCA as well as a theoretical model. While showing small deviations in *p*-values that partly affect the significance levels, the direction and magnitude of all effects can be considered independent of the corresponding model. More detailed information on the methods and results in regard to the robustness tests are made available as supplementary online material.

In order to generate a more comprehensible representation of the results presented in Table 4 and to convey a clearer picture of the success and failure rates as a function of our independent variables *skill* and *pressure*, we classified our data into the four different categories *low skill/high pressure*, *low skill/low pressure*, *high skill/high pressure*, and *high skill/low pressure*. The classification is based on a median split of *pressure* and *skill* in order to obtain groups sizes that are as even as possible for each category. Values higher than or equal to the median are allocated to the

Table 3

Results of binary logistic regressions using the two dimensions *skill* and *pressure* (obtained from the conducted MCA) as independent variables and our three dependent variables *scored*, *missed the goal entirely*, and *saved by the goalkeeper*. Please note that for the variable *saved by the goalkeeper*, we only considered the 1574 penalty kicks that were actually shot on target (i.e., we excluded the ones that missed the goal entirely). For the variables *scored* and *missed the goal entirely*, all 1711 penalty kicks were analyzed.

Dimension	Coefficient	SE	Wald	df	p	Odds Ratio
Scored						
Pressure	−0.11	0.05	3.82	1	.050	0.90
Skill	0.10	0.05	3.12	1	.077	1.10
Constant	1.07	0.06	369.61	1	< .001***	2.91
Missed the Goal Entirely						
Pressure	0.43	0.08	29.04	1	< .001***	1.54
Skill	−0.01	0.09	0.01	1	.909	0.99
Constant	−2.52	0.10	701.68	1	< .001***	0.08
Saved by the Goalkeeper						
Pressure	−0.07	0.07	.97	1	.325	0.94
Skill	−0.13	0.06	4.16	1	.041*	0.88
Constant	−1.45	0.06	502.75	1	< .001***	0.24

Note. **p* < .05. ***p* < .01. ****p* < .001.

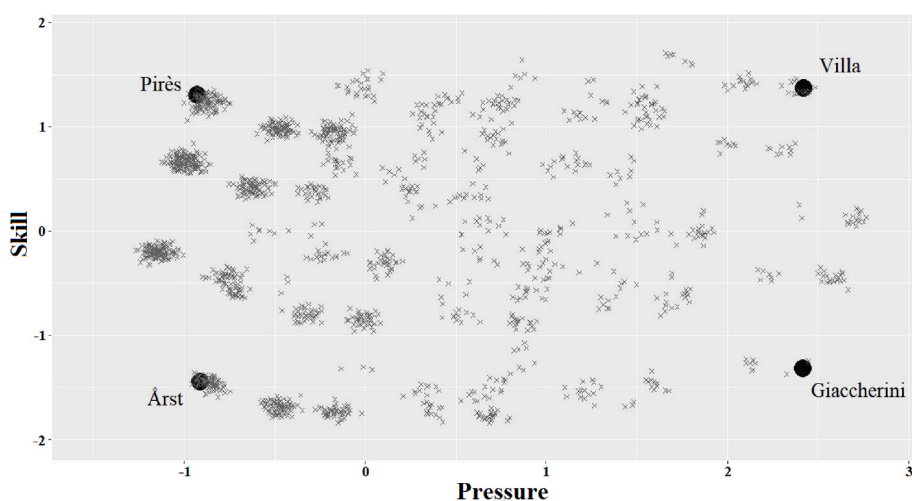


Figure 2. All penalty kicks from our dataset illustrated in a two-dimensional scatterplot, where the x-axis refers to the dimension *pressure* and the y-axis refers to the dimension *skill*. The value of each individual data point reflects the result of the conducted MCA while each grey cross refers to one penalty kick. Four penalty kicks from the dataset are highlighted exemplarily as large black circles. For graphical reasons, the datapoints have been slightly disassembled as otherwise multiple points would overlap at the exact same two-dimensional coordinates due to the categorical data.

Table 4

Percentage of penalties that were *scored*, *missed the goal entirely* or were *saved by the goalkeeper* in regard to the categories *low skill/high pressure*, *low skill/low pressure*, *high skill/high pressure*, and *high skill/low pressure* of our binary classification.

Scored		
	High Pressure	Low Pressure
Low Skill	71.1% (317 of 446)	73.7% (300 of 407)
High Skill	74.2% (317 of 427)	78.4% (338 of 431)
Missed the Goal Entirely		
	High Pressure	Low Pressure
Low Skill	11.2% (50 of 446)	5.4% (22 of 407)
High Skill	10.5% (45 of 427)	4.6% (20 of 431)
Saved by the Goalkeeper		
	High Pressure	Low Pressure
Low Skill	19.9% (79 of 396)	22.1% (85 of 385)
High Skill	17.0% (65 of 382)	17.8% (73 of 411)

category of *high pressure* or *high skill* while values lower than the median are allocated to the category of *low pressure* or *low skill*. Table 4 summarizes the percentages of penalties that were *scored*, *missed the goal entirely*, or were *saved by the goalkeeper* in regard to these four categories.

4. Discussion

The aim of the present research was to investigate the influence of *skill* and *pressure* on the success in penalty kicks as the factor *skill* specifically has not yet been adequately addressed in the existing literature. In accordance with previous research, we assumed that the amount of pressure in the situation in which a penalty kick is taken has an influence on the outcome (e.g., Ellis & Ward, 2022; Jordet et al., 2009; Jordet & Hartman, 2008; Navia et al., 2019). If a player's skill level has a positive impact on the success in penalty kicks, players with higher skill should score more penalty kicks than players with comparably lower skill. In this case, the success rates of highly skilled players in low-pressure situations should be the highest while the success rates of players with a comparably low skill level in high-pressure situations should be the lowest. However, if the results do not show that the skill level has an influence on the success in penalty kicks, this would support the assumptions made in research conducted by Jordet et al. (2007).

Table 1 shows the influence of our six independent variables on our three dependent variables individually. Almost all descriptive results are in line with our expectations. For example, the success rate of the goalkeeper decreases with an increasing market value of the player, fewer penalty kicks are scored in later tournament phases, and fewer kicks are scored in shootouts. As such, the idea that pressure and skill influence the results of penalties is already supported tentatively.

The *team quality* has a significant effect on both the general penalty kick success rate (*scored vs. not scored*) and the number of kicks that miss the goal entirely (*shot on target vs. missed the goal entirely*). That stronger teams scored significantly more penalty kicks and missed the goal significantly less often than weaker teams reflects the finding of Krumer (2020). The result confirms on a more detailed level (we evaluated individual penalty kicks instead of the mere outcomes of whole shootouts) that skill matters in high-pressure situations that involve allegedly simple mechanical tasks such as taking a penalty kick.

The *type of competition*, *tournament phase*, and *type of penalty kick* all have a notable significant effect on the number of kicks that miss the goal entirely (*shot on target vs. missed the goal entirely*). Players miss the goal almost three times more often when they play for their national team compared to when they play for their club team. From the quarterfinal, players miss the goal roughly twice as often as in the stages before. The same applies to shootouts where players miss the goal almost twice as often as when taking an in-game penalty kick. In

accordance with the existing literature (e.g., Ellis & Ward, 2022; Jordet et al., 2009; Jordet & Hartman, 2008; Navia et al., 2019), this result is a clear indicator of the negative influence of pressure on the success in penalty kicks. One possible explanation for this result could be that players get so worried about total failure (i.e., missing the goal entirely) in these high-pressure situations that their thoughts are drawn to this very outcome, a phenomenon that is known as *ironic effects* (also *paradoxical effects* or *ironic processes*, Wegner, 1994). According to Wegner (1994), the deliberate attempt to ignore a thought or action can enhance the inclination to do exactly the opposite, that is, to engage in the very thought or action that is not wanted. Consequently, direct thoughts like “don't kick the ball next to the goal” or indirect ones like “if I miss this penalty kick, my teammates and the fans will be very disappointed with me” can result in exactly what the player did *not* want to do (for related examples in the context of football penalty kicks see Bakker, Oudejans, Binsch, & Van Der Kamp, 2006; Binsch, Oudejans, Bakker, Hoozemans, & Savelsbergh, 2010; Binsch, Oudejans, Bakker, & Savelsbergh, 2010; Navarro, van der Kamp, Ranvaud, & Savelsbergh, 2013). Furthermore, it is known that high levels of pressure and anxiety can have a detrimental impact on cognitive performance (Eysenck, Derakshan, Santos, & Calvo, 2007), which can account for changes in the coordination and in the regulation of movements. This, in turn, can result in decrements of athletic performance. The well-known phenomenon *choking under pressure* (Beilock & Carr, 2001; Beilock, Carr, MacMahon, & Starkes, 2002), for example, can be observed when individuals fail to successfully accomplish a well-trained and easy task in a situation that is highly stressful (e.g., a penalty kick; see Brimmell et al., 2018; Dohmen, 2008; Ellis & Ward, 2022; Furley et al., 2020; Jordet, 2009a; Jordet et al., 2012; Memmert et al., 2013; Navarro et al., 2012; Navia et al., 2019; Wood, 2010). The existing literature in the field of general psychology shows that anxiety is associated with an attentional bias towards the processing of threat-related distracters (and the inability to then disengage from these distracters). Also, anxiety can lead to an increased distractibility when task-irrelevant threatening stimuli are present (Wood & Wilson, 2010a; for a review, see Bar-Haim, Lamy, Pergamin, Bakermans-Kranenburg, & Van Ijzendoorn, 2007). A framework that aims to encompass these attentional disruptions is Attentional Control Theory (ACT; Eysenck et al., 2007). The primary assumption of ACT is that attention is controlled by two systems—a goal-driven system (top-down) that is influenced by current goals, expectations, and knowledge; and a stimulus-driven system (bottom-up) that is influenced by salient or conspicuous stimuli (Corbetta & Shulman, 2002). If the balance between these two attentional systems gets disrupted by anxiety in favor of the stimulus-driven system, this could have a negative impact on athletic performance (especially in high-pressure situations such as penalty shootouts in prestigious competitions; for further literature that investigates the performance in penalty kicks in consideration of ACT, see Brimmell et al., 2021; Navarro et al., 2012; Navarro, Miyamoto, et al., 2013; Noel & Van Der Kamp, 2012; Wilson, Wood, & Vine, 2009; Wood, 2010).

The *type of competition* also has a significant impact on the goalkeeper success rate (*not saved by the goalkeeper vs. saved by the goalkeeper*). As the general success rate is nearly unaffected by the type of competition but the number of kicks that miss the goal entirely is significantly higher for national teams, this logically results in a higher success rate for goalkeepers in the UEFA Europa League and UEFA Champions League. That goalkeepers save significantly more penalty kicks in games involving club teams than in games involving national teams can therefore be partly explained by the consideration presented above. A further possible explanation is that teams who play in the major tournaments on a club-level are able to employ the very best goalkeepers in the world (i.e., the choice of the goalkeeper is independent of his nationality) whereas national teams are obligated to play goalkeepers with the corresponding nationality. Especially smaller football nations may be forced to play goalkeepers with a comparably lower skill level, specifically in the FIFA World Cup (here, the number of countries with a lower

national team ranking is higher compared to the UEFA European Championship; see <https://www.fifa.com/de/fifa-world-ranking/men>). This tentatively supports the descriptive result that the general success rate of a player decreases with an increasing market value of the goalkeeper (even though this finding was non-significant) and, besides that, generally speaks for a high goalkeeper skill level in major tournaments on a club-level.

In addition to these expectable results, however, we also found some surprising exceptions of which the first one involves the influence of the *player market value* on the number of kicks that missed the goal entirely (*shot on target vs. missed the goal entirely*): While penalty kicks by players with a high market value were scored most often and were least often saved by the goalkeeper, these players unexpectedly show the highest rate of kicks that missed the goal entirely (9.6% compared to 7.3% and 7.1% for players with middle and low market values respectively). A possible explanation for this result is that less skilled players try to take less risk in order to avoid total failure (i.e., missing the goal entirely) by not kicking the ball too close to the goal post. This in turn results in a higher success rate for the goalkeeper. Players with a high skill level, on the other hand, seem to be willing to take the more difficult shot and thus more risk at the same time in order to maximize their chances, even if it means that the probability of missing the goal entirely increases—a strategy that seems to pay off. In the context of football penalty kicks, there is no empirical evidence yet to support the assumption that taking the more difficult shot goes hand in hand with an increased success probability. However, there is research examining the relationship between risk-taking and reward in basketball (e.g., Williams, Nesbitt, Eidels, & Elliott, 2011) that points in a similar direction. The 3-point shot in basketball was introduced in 1961 to reward riskier attempts (i.e., more difficult shots) that are taken further away from the basket (Nourayi, 2020). Studies that evaluated the shooting performance of professional basketball players since then state that 3-point shots offer higher returns (higher expected points per shot) and therefore offset the difficulty when taken by a player with a high shot efficiency (e.g., Sandholtz, Mortensen, & Bornn, 2020).

The influence of the *goalkeeper market value* on the success of the penalty kick taker is equally surprising. As expected, penalty kick takers missed the goal entirely significantly more often against goalkeepers with a high market value. Also, penalty kicks were scored most often against goalkeepers with a low market value on a descriptive level (success rates of 72.3%, 74.6%, and 75.8% against goalkeepers with low, middle, and high market values respectively), however, this result was not significant. What is most surprising is that the probability to save a penalty that is shot on target is basically identical for all goalkeepers. A possible explanation for these results is that goalkeepers with a higher market value may be perceived as a bigger threat. This could lead players to feel intimidated and wanting to kick the ball as precise (and as far away from the goalkeeper) as possible, resulting in a higher chance of missing the goal entirely. A study conducted by Masters et al., 2010 provides potential evidence for our explanatory approach as they were able to show that goalkeepers with a high status or reputation are perceived bigger in size than they actually are. Further related indications on the influence of visual illusions on the kicking behavior of a player who takes a penalty can be found in van der Kamp and Masters (2008). Moreover, there are findings from the field of economics that hint in a similar direction. According to Brown (2011), the mere presence of a highly skilled individual can be associated with a drop in performance of others involved in the situation, a phenomenon that is coined the *adverse superstar effect*.

As expected, the conducted Spearman correlations show that the majority of our independent variables are highly intercorrelated. This indicates that an individual evaluation of these variables is not sufficient and that, consequently, an appropriate component reduction method such as an MCA is required (Kamalja & Khangar, 2017). The outcome of the MCA (Table 2) shows the statistical reduction of our six independent variables to two dimensions. Since it is only possible to match these two

dimensions to our target concepts *skill* and *pressure* by argument, we examined the values obtained from the MCA in the context of our entire data set. The four examples of penalty kicks taken by different players in varying contexts illustrated in Figure 2 show that the components found in the MCA can indeed be interpreted as reasonable reflections of the situational pressure and the general skill of the player who takes the penalty kick.

We, therefore, used the two dimensions *skill* and *pressure* obtained from the MCA as independent variables in binary logistic regression models for further analyses. The results presented in Table 3 show that players missed the goal entirely significantly more often when the pressure was high, while this seems to be largely independent of the factor of skill. This result clearly reflects the phenomenon “choking under pressure” (Beilock & Carr, 2001; Beilock et al., 2002). One of the most prominent examples in this regard is the penalty kick taken by Bastian Schweinsteiger in the 2012 UEFA Champions League final between FC Bayern München and FC Chelsea. Schweinsteiger, who was considered one of the world’s best football players at the time, took the decisive kick and failed.

The reverse effect can be found for penalty kicks that were saved by the goalkeeper. The probability that a kick was saved was not affected by pressure, but goalkeepers saved significantly fewer penalties when they were kicked by players with a high skill level. In line with these findings, the descriptive probability that a penalty kick was scored decreased with higher pressure and increased with higher skill, however, both results closely fail to be significant at a 5% level. These results reflect the predictions from the field of economics, according to which “individuals with greater skill are supposed to enhance their performance when a lot is at stake” (Cohen-Zada et al., 2018; González-Díaz et al., 2012; Iqbal & Krumer, 2019; Jetter & Walker, 2015; Lazear, 2000; Rosen, 1986). The descriptive results illustrated in Table 4 mirror these exact findings and their respective interpretation in a more comprehensible and intuitive way.

In summary, we come to the following conclusion: High situational pressure increases the probability of missing the goal entirely, independent of the player’s skill level. This is by far the clearest result of our analysis. Also, the probability that the goalkeeper saves a penalty that is shot on target decreases when a highly skilled player takes the kick. In general, high situational pressure decreases the probability of scoring a penalty kick as shown before in past research (e.g., Ellis & Ward, 2022; Jordet et al., 2009; Jordet & Hartman, 2008; Navia et al., 2019). Furthermore, in contrast to the findings by Jordet et al. (2007), the probability to score a penalty increases if a highly skilled player takes the kick. This indicates that a high skill level can indeed act as a kind of buffer against debilitating effects caused by performance pressure (see Cohen-Zada et al., 2018; Ehrenberg & Bognanno, 1990; Iqbal & Krumer, 2019; Jetter & Walker, 2015; Lazear, 2000; Rosen, 1986), also in football penalty kicks.

Author contributions

M.B. and F.W. developed the concept of the study. Data collection was performed by M.B. The data analysis as well as the graphical illustration of results was performed by F.W. and M.B. All authors contributed to the interpretation of the results. M.B., F.W., and P.F. drafted the manuscript, D.M. provided critical revisions. All authors approved the final version of the manuscript for submission.

Availability of materials and data

The dataset generated and analyzed during the current study will be made available by the corresponding author upon reasonable request.

Declaration of conflicting interests

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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